MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Restricted Antimicrobials Requiring AST Approval in Patients on Adult Services

Use of certain antimicrobial agents is restricted at Michigan Medicine. Agents are classified as Tier I or Tier II agents depending on whether Antimicrobial Stewardship Team (AST) approval is required prior to dispensing.

TIER I RESTRICTED ANTIMICROBIALS

The use of the following agents (i.e., Tier I agents) requires AST (pager #30780) or ID approval prior to dispensing between the hours of 0700 and 2300. Please consult appropriate treatment guidelines.

Note: The below indications generally refer to appropriate EMPIRIC use. When cultures are available, antibiotic therapy should be escalated/de-escalated as appropriate based on organism and susceptibility. Restricted agents should only be utilized if narrower-spectrum agents are resistant or otherwise inappropriate. When cultures are not available, please refer to individual treatment guidelines for appropriate definitive therapy strategies.

UMHHC Policy 07-01-015 ("Use of Infectious Diseases Restricted Antimicrobials")
All treatment guidelines are available on the Antimicrobial Stewardship page

Restricted Medication	Approved Reasons for Use
Aztreonam-avibactam	● For infections due to MBL-producing (NDM, VIM, or IMP+) Enterobacterales or Stenotrophomonas (alternative therapy in serious infections).
Baloxavir	● As per Influenza Treatment and Prophylaxis Guidelines. Single-dose therapy for influenza B infection does not require approval.
Cefiderocol	● Generally, for difficult-to-treat/extremely drug resistant (DTR/XDR) gram-negative infections. Bloodstream infection treatment guidelines provide recommendations on specific microbiological scenarios for use.
Ceftazidime-avibactam	● Generally, for DTR/XDR gram-negative infections. Bloodstream infection treatment guidelines provide microbiological recommendations on specific scenarios for use.
Ceftaroline	● Treatment of Staphylococcus aureus infections ○ Alternative therapy to vancomycin and daptomycin, depending on specific. May also be used in combination.
Ceftolozane-tazobactam	● Treatment of DTR Pseudomonas aeruginosa infections.
Dalbavancin	● Patients who cannot be discharged with PICC vancomycin or daptomycin due to IV drug use, lack of access, psychological or social issues ● Oral therapy is not option due to resistance, intolerance, or required duration therapy is > 2 weeks ● Patients with psychological or social issues limiting the ability to administer IV antibiotics ● Patients that require discharge to SAR or nursing home for the sole purpose of administering IV antibiotics ● Patients with a history of frequent discharges against medical advice
Daptomycin	● Surgical prophylaxis as per "Surgical Antimicrobial Prophylaxis Guidelines" ● Treatment option for infections due to Enterococcus and/or Staphylococcus.
Eravacycline	● Alternative to tigecycline when tigecycline not tolerated.
Ertapenem	● Empiric or definitive therapy of infections caused by Gram negative infections when oral or other once-daily intravenous antibiotics are not preferred ● Treatment for refractory hidradenitis suppurativa in consultation with dermatology and infectious diseases when alternative therapies are not possible ● Combination therapy with cefazolin for refractory methicillin-susceptible *Staphylococcus aureus* infections ● Surgical prophylaxis for patients colonized or infected with ESBL and/or AmpC-producing organisms
Ethanol Lock Therapy	● There are extensive exclusion criteria. See complete guideline on Antimicrobial Stewardship website for full criteria for use.

Restricted Medication	Approved Reasons for Use
Fidaxomicin	● As defined in C difficile Treatment Guidelines.
Imipenem	● Patients with a infection due to a gram-negative when meropenem is not an option ● Patients with infections due to *Nocardia spp*.
Imipenem-Relebactam	● Treatment of DTR Pseudomonas aeruginosa infections
Isavuconazole	● Treatment of aspergillosis and mucormycosis as per those treatment guidelines. Alternative for treatment of other invasive infections due to filamentous molds. ● Prophylaxis in settings where other mold-active alternatives are limited by drug-drug interactions or safety considerations.
Letermovir	● Prophylaxis in CMV-seropositive allogeneic HSCT recipients as per BMT prophylaxis guidelines. ● Prophylaxis in other populations in which standard approaches (typically valganciclovir) are not tenable due to resistance and/or tolerability concerns. ● Treatment of CMV disease in cases recommended by consultation with Transplant Infectious Diseases
Linezolid	● Per Treatment Guidelines ● Treatment option for infections due to gram-positive organisms.
Liposomal Amphotericin B	● Treatment of invasive fungal infections as per treatment guidelines.
Maribavir	● Treatment of CMV in adult and pediatric patients (12 years of age and older and weighing at least 35 kg) with CMV disease that is refractory to treatment (with or without genotypic resistance) with ganciclovir, valganciclovir, cidofovir or foscarnet or when these agents are not possible due to toxicity and/or allergies.
Meropenem	● As per treatment guidelines. ● Treatment of infections caused by organisms not able to be treated with piperacillin/tazobactam and cefepime
Meropenem-vaborbactam	● Treatment of highly suspected or documented extensively drug-resistant gram-negative pathogens (ex. KPC-producing Enterobacteriaceae). ● Meropenem-vaborbactam is the preferred treatment for infections caused by KPC-producing Enterobacteriaceae
Micafungin	● As per Treatment and Prophylaxis guidelines
Minocycline IV	● Treatment of documented Acinetobacter, Stenotrophomonas, or other non-lactose fermenters and the patient is unable to take oral minocycline
Nirmatrelvir- Ritonavir	● COVID-19 treatment in B cell depleted patients.
Omadacycline	● Nontuberculosis mycobacterial infection ● Treatment of non-urinary tract, vancomycin-resistant enterococci infections
Peramivir	● Inpatients with influenza in which drug delivery by a route other than IV is not feasible
Posaconazole	● As per treatment and prophylaxis guidelines ● Treatment of invasive fungal infections.
Rezafungin	● Treatment of invasive candidiasis in the following scenarios: ○ Patients unable to take oral azoles / oral therapy is not deemed appropriate ○ 1x dose prior to discharge is an option for patient unable to take oral azoles that have ≤ 7 days of therapy remaining ○ Patients unable to maintain IV access outpatient or when IV access weighs potential risk

Restricted Medication	Approved Reasons for Use
Inhaled Ribavirin	• Restricted to UMHS guidelines for use entitled "Indications for the Use of Ribavirin for Treatment of Respiratory Viral Infections in Adult and Pediatric Patients" • ID consult is mandatory prior to starting therapy
Sulbactam-durlobactam	• Treatment of serious infection due to *Acinetobacter* that is resistant to ampicillin-sulbactam and carbapenems. • Treatment of other XDR gram-negative infections.
Taurolidine/Heparin (Defencath)	• Primary Prevention- o UMHS nephrology clinics for HD patients. o Consideration in pediatric patients with any central line meeting current criteria for ethanol lock therapy o Continuation from the outpatient setting for adult patients with any central line • Secondary Prevention- o High-risk patients with any central line with pathogen resistance or intolerance of standard lock strategies (e.g., heparin, ethanol, or antibiotic locks), and one of the following:, including those with a history of recurrent line infection, inability to remove or replace the catheter due to limited access, or failure. o Continuation from the outpatient setting for adult patients with any central line • Active treatment of CRBS • Treatment of central line infections when trying to salvage the line in patients with any central line that are not candidates for alternate antimicrobial lock therapy.
Tigecycline	• Treatment of XDR gram-negative infections • Treatment of vancomycin-resistant enterococci infections resistant to daptomycin and linezolid, or patient intolerance to daptomycin and linezolid • Empiric therapy for patients with significant allergies to alternative agents
Voriconazole	• As per treatment and prophylaxis guidelines. • Treatment and prophylaxis of other invasive fungal infections.

Last revised: 02/2026	**Antimicrobial Stewardship Subcommittee:** 02/2026	**P&T:** 03/2026

Revision History	
Date	**Revision**
01/2020	Added eravacycline
08/2020	Added cefiderocol and imipenem-relebactam
11/2020	Added remdesivir
04/2022	Removed moxifloxacin, updated ertapenem
11/2022	Added omadacycline
06/2023	Updated posaconazole and isavuconazole
03/2024	Added rezafungin, sulbactam-durlobactam, baricitinib
02/2026	Comprehensive updates for simplicity and incorporating system-wide considerations
09/2026	Added Maribavir and Taurolidine/Heparin